

Lecture 1 — Foundations of Engineering Ethics

1. What Is Engineering Ethics?

Engineering ethics is the study of how engineers should make professional decisions that affect society, especially when safety, uncertainty, and responsibility are involved.

- Professional judgment
- Public safety
- Responsibility
- Risk
- Decision-making

2. Ethics vs. Engineering Ethics

Ethics	Engineering Ethics
General moral behavior	Moral decisions in engineering work
Personal life	Professional life
Small-scale impact	Large-scale public impact

★EXAM LINE *Engineering ethics begins when a technical decision affects other people.*

3. Why Engineering Ethics Matters

Engineering decisions can:

- Save lives
- Injure people
- Kill people
- Affect society for years

Failures are often caused by:

- Poor communication
- Management decisions
- Schedule pressure
- Ignoring warnings

...not only bad engineering.

★EXAM LINE *Engineering failures are usually both technical and ethical failures.*

4. Ford Pinto Case

- Fuel tank defect was already known.
- Company continued selling cars.
- Rear-end collisions caused fires.
- People died.

LESSON *Knowing about a danger and still releasing the product is unethical. The issue is not the accident itself — the issue is continuing despite knowing the risk.*

5. Challenger Disaster

- Engineers warned about O-rings.
- Cold weather increased failure risk.
- Management ignored warnings.
- Shuttle exploded.

LESSON *Ignoring expert safety advice can lead to disaster.*

6. Columbia Disaster

- Foam had hit the shuttle before — nothing happened.
- People assumed: "It must be safe."
- Eventually one mission failed.

LESSON *Past success does not prove future safety. Repeated survival creates false confidence.*

7. Moral Autonomy

An engineer should use independent professional judgment instead of blindly following orders.

Boss says	Engineer knows	Professional ethics says
"Ship the product."	It isn't safe.	Speak up.

8. Good Intentions Are Not Enough

People don't usually intend to cause disasters. Failures happen because of:

- Pressure
- Incomplete information
- Poor judgment
- Normalized risk



9. Managing Uncertainty

Good engineers ask:

- What if we're wrong?

- Who is affected?
- Have we tested enough?
- What happens if this fails?

Good practice:

- Test thoroughly
- Document uncertainty
- Expect failures
- Don't pretend to know everything

10. Burden of Risk

If evidence is incomplete, who should bear the risk?

Option	Who bears the risk / cost
Option A: Release now.	Public bears the risk.
Option B: Delay release.	Company bears the cost.

KEY RULE *Engineering ethics usually favors protecting the public.*

11. Personal Ethics vs. Professional Ethics

Professional ethics involves:

- Employers
- Clients
- Regulators
- Society

Consequences are much larger than in personal ethics.

12. Ethics vs. Law ★Favorite exam topic

★EXAM LINE *Legal ≠ Ethical*

- **Legal but unethical:** Selling a dangerous product that technically meets regulations.
- **Illegal but ethical:** Breaking an outdated law to prevent serious harm (context matters).

★EXAM LINE *Law sets the minimum standard; ethics often requires higher standards.*

13. Ethical Problems Are Like Design Problems

Both involve:

- Trade-offs
- Uncertainty
- Incomplete information
- Judgment

But uncertainty does not mean “anything is acceptable.”

14. If Management Ignores You

Professional responsibility, in order:

1. Document concerns.
2. Report through proper channels.
3. Refuse participation if necessary.
4. External disclosure (whistleblowing) — only if serious risk remains and internal methods fail.

★High-Value Exam Lines — Lecture 1

- Engineering ethics begins when technical decisions affect other people.
- Engineers have a responsibility to protect public safety above personal or organizational interests.
- Good intentions cannot replace good professional judgment.
- Law provides the minimum standard; ethics often demands more.
- Uncertainty increases the need for careful judgment; it does not remove responsibility.
- Past success does not prove a design is safe.
- Professional engineers must exercise moral autonomy and not blindly obey authority.

Cases to Remember — Lecture 1

Case	What Happened	Lesson
Ford Pinto	Known fuel tank defect	Don't ignore known hazards
Challenger	O-ring warnings ignored	Listen to engineers and prioritize safety
Columbia	Previous foam strikes normalized	Don't mistake repeated success for safety

Lecture 2 — Engineering as a Profession

1. Job vs. Occupation vs. Profession

Job	Occupation	Profession
Any paid work	Regular field of work	Occupation with special requirements
No special qualifications	Makes a living	Requires education, ethics, judgment, and public service

★EXAM LINE *Every profession is an occupation, but not every occupation is a profession.*

2. Four Attributes of a Profession ★★★

1. Skills, Judgment & Discretion

- Specialized knowledge
- Independent decision-making
- Confidentiality
- Decisions affect people's lives

2. Extensive Formal Education

- Requires structured education
- Not just experience or apprenticeship

3. Professional Societies

- IEEE, NSPE, ASME

Purpose: set standards, promote ethics, regulate professional conduct.

4. Significant Public Good

Profession benefits society.

3. Why Engineering Is a Profession

Engineering satisfies all four attributes:

- ✓Sophisticated skills & judgment
- ✓Formal education
- ✓Professional societies
- ✓Public good

BUT *Engineering is less mature than medicine and law because engineers mostly work for companies, licensing is often optional, and engineering societies are weaker.*

4. Judgment vs. Discretion

Judgment: Making important decisions using knowledge and experience. Example: choosing bridge material.

Discretion has two meanings:

- **Confidentiality:** Keeping client/employer information secret.
- **Autonomy:** Freedom to make professional decisions.

5. Professional Societies

IEEE, ASME, ASCE, AICE, IIE, etc.

6. Social Contract Model vs. Business Model

Social Contract Model	Business Model
Profession exists to serve society.	Profession exists mainly to increase salaries.
Society gives professionals: respect, autonomy, high status.	Reduce competition.
Professionals give society: ethical service, competence, public protection.	Protect members.

★**EXAM LINE** *Engineering mostly follows the Social Contract Model.*

7. Code of Ethics

A code of ethics explains how moral principles apply to professional practice.

- ✓Framework for ethical decisions
- ✓Starting point
- ✓NOT a law
- ✓Doesn't create new morals
- ✓Applies existing morals

8. Purposes of a Code of Ethics

- Creates an ethical workplace
- Supports employees refusing unethical orders
- Shows professionalism
- Builds public trust

9. Limitations of a Code of Ethics

Three objections:

5. Engineers may not belong to professional societies.
6. Many engineers don't read the code.
7. Rules inside the code may conflict.

10. Evolution of Engineering Ethics

Time	Focus
Late 19th century	No formal code
Early 20th century	Business conduct

Time	Focus
Mid 20th century	Employer duties
Modern	Public safety, health, environment

11. IEEE vs. NSPE

IEEE	NSPE
Short	Long
General	Detailed
Easy to read	Harder to read
Mentions environment	Doesn't explicitly mention environment
Doesn't discuss employer duty	Discusses employer duty

12. Most Important Rule ★★★★★

KEY RULE *If employer instructions conflict with public safety, Public Safety Comes First. This overrides employer loyalty and is one of the highest-priority principles in engineering ethics.*

13. BART Case

- Engineers reported unsafe train systems.
- They were fired.
- IEEE supported them.
- Engineers still lost.

LESSON *Engineering societies have limited power to protect whistleblowers.*

14. Case Studies

Intel Pentium

Problem: Intel knew the processor had a defect and continued selling it.

Ethics: Lack of transparency, misleading customers, loss of trust.

Denver Airport Runway

Problem: Concrete was diluted; test reports were falsified; inspectors were deceived.

Ethics: Dishonesty, fake data, public safety endangered.

Paradyne Computer

Problem: Bid using a non-existent computer; fake demonstration; relaxed acceptance tests.

Ethics: Misrepresentation, deception, conflict of interest.

High-Value One-Liners — Lecture 2

- Engineering is a profession because it satisfies all four professional attributes.
- Professional judgment means making important decisions using knowledge and experience.

- Discretion means confidentiality and professional autonomy.
- Codes of ethics are guidelines, not laws.
- Codes do not create morality; they apply existing moral principles.
- Public safety is the highest responsibility of an engineer.
- Engineering societies help maintain professional standards and ethical conduct.
- Engineering is less mature than medicine and law due to weaker licensing and professional regulation.
- Ethical failures often involve dishonesty, concealment, or putting employer interests above public safety.

Lecture 3 — Ethical Theories

1. Main Theme

★EXAM LINE *Engineering ethics is about making morally correct decisions, not just technically correct ones.*

Engineers should consider:

- Consequences
- Duties
- Rights
- Character
- Public safety

2. Bhopal Disaster ★★★★★Very important case



Why it matters: the disaster was not just a technical failure. It became an ethical failure because people ignored warnings, reduced maintenance, disabled safety systems, and failed emergency planning.

★EXAM LINE *Technical failure became moral failure.*

Engineering responsibility includes: maintenance, safety systems, emergency planning, protecting the public.

3. Ethical Theories

Engineering asks: Can we build it?

Ethics asks: Should we build it? Is it fair? Is it safe? Is it morally acceptable?

Both help solve problems systematically.

4. Four Ethical Theories ★★★★★Most important

Theory	Main Question	Focus	Keywords
Utilitarianism	Which action gives the greatest overall good?	Consequences	Benefit, cost, welfare
Duty Ethics	What duties must I obey?	Obligations	Honesty, fairness, responsibility
Rights Ethics	Whose rights must be protected?	Individual rights	Life, safety, privacy, consent

Theory	Main Question	Focus	Keywords
Virtue Ethics	What kind of engineer am I becoming?	Character	Honesty, competence, trustworthiness

★**EXAM LINE** *These are lenses, not competing answers. Use all four before making a decision.*

5. Utilitarianism

Goal: maximize total welfare.

Engineering use: risk-benefit analysis, cost-benefit analysis.

PROBLEM *It may ignore people who suffer. Millions may benefit while a small community suffers greatly — the theory may still approve the project.*

Always ask: Who benefits? Who pays? Who is ignored?

6. Cost-Benefit Analysis (CBA)

Good for comparing: cost, feasibility, expected gains.

Cannot measure well: dignity, fairness, trust, fear, environmental damage.

★**EXAM LINE** *Cost-benefit analysis informs ethical judgment but does not replace ethical judgment.*

Decision rule: (1) find ethical options, (2) then compare cost.

7. Duty Ethics

Main duties:

- Tell the truth.
- Avoid preventable harm.
- Be fair.
- Respect consent.

Example: hiding a dangerous defect is wrong even if it saves money.

8. Rights Ethics

- Life
- Safety
- Property
- Liberty
- Privacy
- Informed consent

★**EXAM LINE** *A large public benefit does not automatically justify harming a few people.*

9. Virtue Ethics

Good engineer qualities:

- Honesty

- Competence
- Responsibility
- Fairness
- Trustworthiness

Ask yourself: Is this honest? Does it show competence? Would I be proud if everyone knew about this decision?

10. Dangerous Virtue Words

These words sound good but can become unethical:

- Loyalty
- Honor
- Pride
- Team spirit

WRONG *"I stayed loyal by hiding the accident." — Professional loyalty is first to public safety and honesty.*

11. Corporate Responsibility



The company is accountable. Individuals are also accountable.

DON'T SAY *"The system made me do it."*

12. Which Theory Should Be Used?

Five-step method:

8. State the ethical problem.
9. Analyze using all four theories.
10. Compare agreements and conflicts.
11. Give serious weight to duties and rights.
12. Make a balanced judgment.

13. Bhopal Through Four Theories

- **Utilitarianism:** Economic benefit cannot justify catastrophic risk.
- **Duty Ethics:** Company failed maintenance, warning, and emergency planning.
- **Rights Ethics:** People's right to safety was violated.
- **Virtue Ethics:** Company lacked honesty, competence, and diligence.
- **Corporate Morality:** Responsibility belongs to engineers, managers, designers, and regulators.

14. Ethics Across Cultures

Different traditions: Western, Chinese, Indian, Muslim, Buddhist — different philosophies, same engineering values.

Common values: honesty, competence, safety, fairness, respect.

★**EXAM LINE** *Engineering ethics is universal because engineering risks are universal.*

15. Ethical Traditions Easy marks

Tradition	Core Values	Engineering Application
Chinese	Character, respect, community	Balance society and individuals
Indian	Truth, compassion, self-control, avoid greed	Think beyond profit
Muslim	Honesty, humility, trustworthiness	Truthful claims and fair dealing
Buddhist	Avoid harming life, lying, stealing; promote safety	Sustainability and competence

16. Aberdeen Three Case

- Leaking chemicals
- Poor storage
- Corrosion
- Pollution

Managers were convicted.

LESSON *Authority creates accountability. Managers are responsible because they had authority over safety.*

17. Aberdeen Analysis

- **Utilitarianism:** Unsafe storage created huge risk with little benefit.
- **Duty Ethics:** Managers had duties to control hazardous materials.
- **Rights Ethics:** Workers had the right to safety.
- **Virtue Ethics:** “We’ve always done it this way” is not an ethical excuse.

18. AI Example

Automated welfare system —

Benefits: faster, cheaper.

Problems: more mistakes for some people, unfair treatment, people cannot easily challenge AI decisions.

★**EXAM LINE** *Efficiency ≠ Fairness.*

19. Team Ethics

Bad teamwork: hiding mistakes, missing work, hiding delays, refusing to share information.

Good teamwork: reliability, openness, fairness, shared responsibility, helping teammates learn.

★High-Value Exam Quotes — Lecture 3

- Technical failure became moral failure.
- Cost-benefit analysis informs judgment; it does not replace judgment.
- Authority creates accountability.
- Different ethical theories are lenses, not votes.
- Engineering ethics is universal because engineering risks are universal.
- Public safety comes before economic benefit.

Lecture 4 — Ethical Problem-Solving Techniques

Why Is a Method Needed?

Engineering problems have formulas. Ethical problems do not, because:

- Facts may be incomplete.
- Definitions may be disputed.
- Ethical duties may conflict.

★EXAM LINE Ethical methods do not produce the correct answer. They help make decisions that are clear, honest, and justifiable.

Overall Problem-Solving Process



Always follow this order.

Step 1: Identify the Type of Issue

1. Factual Issue

Question: What do we actually know?

Solved using: evidence, research, measurements, records.

Examples: failure rate, safety test results, risk level.

2. Conceptual Issue

Question: What should this action be called?

Solved using: definitions, classifications, comparisons.

Examples: Gift or bribe? Marketing or deception? Existing system or prototype?

3. Moral Issue

Question: Which ethical principle applies?

Situation	Principle
Lying	Honesty
Dangerous design	Safety

Situation	Principle
Unfair influence	Integrity
Unequal treatment	Fairness

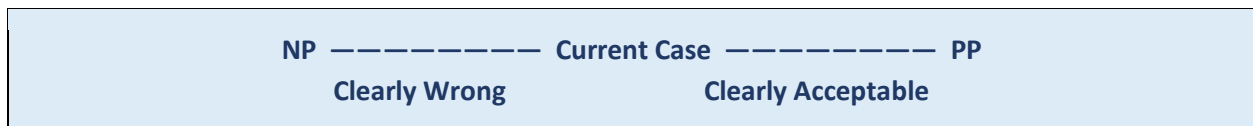
★EXAM LINE Many ethical disagreements are actually factual or conceptual disagreements. Solve those first before making a moral judgment.

Line Drawing Method

Purpose: used when right and wrong are not clearly separated — the case lies in a gray area.

Steps:

13. Choose a clearly wrong case (Negative Paradigm, NP).
14. Choose a clearly acceptable case (Positive Paradigm, PP).
15. Compare similar cases.
16. Place the current case honestly between NP and PP.



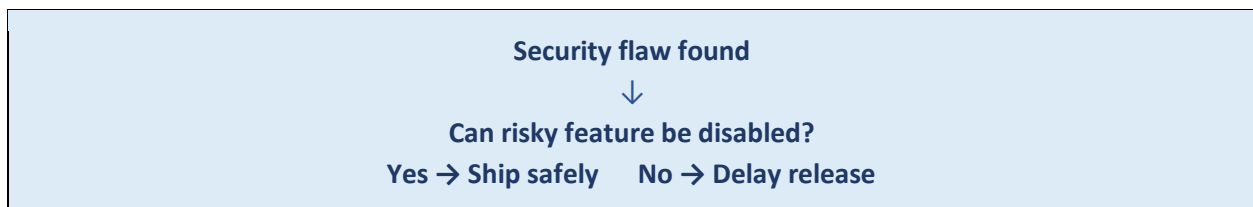
Advantages: helps compare cases, shows how acceptable a decision is, reveals missing information.

LIMITATIONS Line Drawing can fail if examples are biased, facts are hidden, placement is dishonest, or it is treated as absolute proof. It is an aid, not proof.

Flow Charting

Purpose: shows sequence of decisions, consequences, and alternative actions.

Unlike Line Drawing (which compares cases), a Flow Chart compares consequences.



Advantages: makes consequences visible, identifies decision points, prevents unsafe decisions.

Resolving Conflicting Values

Three approaches:

1. Easy Priority

One value clearly outweighs another. Example: Public safety > Company profit.

2. Creative Middle

Find another solution. Example: instead of polluting — reduce waste, treat waste, redesign system.

3. Hard Choice

When no perfect solution exists, choose the most ethically defensible option based on available information.

Major Case Studies — Lecture 4

1. Paradyne Computers

Issue: government requested an existing computer system; Paradyne demonstrated an unfinished system.

- **Factual:** System wasn't finished.
- **Conceptual:** Sales practice or deception?
- **Moral:** Deception is unethical.

LESSON *Don't judge morality before verifying facts and definitions.*

2. Pentium Chip

Issue: Intel knew about a processor flaw. Possible responses: hide problem, replace only if customer complains, inform customers, recall all chips.

LESSON *Greater transparency and customer protection are ethically stronger.*

3. Bhopal Disaster

Flow charts show the accident resulted from many connected decisions: plant location, weak safety standards, maintenance, backup systems, emergency planning.

★EXAM LINE *Weak local laws do not remove an engineer's duty to ensure safety.*

4. Challenger Disaster

Conflict: launch pressure vs. human safety.

Reasons to launch: schedule, contracts, NASA relationship.

Reasons not to launch: O-ring failure, crew safety, uncertain data.

LESSON *Human life has higher priority than schedule, money, or contracts.*

5. Gifts vs. Bribes

Gift — usually acceptable if: small value, transparent, no influence on decisions.

Bribe: a benefit offered to influence dishonest action. Bribery distorts competition, damages trust, and is illegal.

Boundary depends on: value, timing, intent, secrecy, decision-making authority, repeated gifts.

6. Spiro Agnew Kickbacks

Issue: companies paid money to secure government contracts.

- **Factual:** Payments occurred.
- **Conceptual:** Gift, campaign support, or kickback?
- **Moral:** Destroys fairness and public trust.

7. Cell Phones and Cancer

Scientific evidence was uncertain, but engineers still had to design products.

★EXAM LINE *Reduce possible risks whenever reasonable, even if science has not reached a final conclusion.*

Important Definitions — Lecture 4

Negative Paradigm (NP): *A clearly unethical case used as the left endpoint in Line Drawing.*

Positive Paradigm (PP): *A clearly ethical case used as the right endpoint.*

Factual Issue: *Question about evidence and reality.*

Conceptual Issue: *Question about meaning or classification.*

Moral Issue: *Question about ethical principles.*

Line Drawing: *Method for evaluating gray-area ethical cases by comparing them with clearly acceptable and unacceptable examples.*

Flow Charting: *Method for analyzing consequences by following different decision paths.*

Lecture 5 — Professional Responsibilities and Rights

Main Idea

Engineers have both Responsibilities (things they must do) and Rights (things they are entitled to). Both exist to ensure public safety.

BART Case — Full Timeline

17. Engineers found safety flaws.
18. Management ignored them.
19. Anonymous memo sent.
20. Contacted board member.
21. Fired.
22. Lawsuit settled (\$25,000 each).
23. Train overshot station (proved engineers were right).

LESSON *Engineers must protect public safety. Employers should not punish engineers for reporting safety concerns.*

Professional Responsibilities

1. Confidentiality

Keep employer/client information secret. Examples: test results, designs, supplier identities, marketing strategies, production costs.

- Confidentiality continues even after leaving the company.
- Engineers must balance career mobility with protection of company intellectual property (IP).
- Responsibility lies with the engineer.

2. Conflict of Interest

Definition: personal interests interfere (or appear to interfere) with professional judgment.

- **Actual:** Conflict exists now. Example: awarding a contract to your relative.
- **Potential:** Could become a conflict later. Example: owning shares in a bidding company.
- **Apparent:** Looks like a conflict even if none exists. Example: people believe you're biased.

Solutions: disclose, recuse yourself, seek guidance.

3. Competitive Bidding

Companies compete using price proposals.

Ethical problems: lowballing bids, exaggerating qualifications, disparaging competitors.

★EXAM LINE *Compete honestly.*

Professional Rights

General Rights

- Privacy
- Participate in outside activities

- Object to company policies
- Due process (fair treatment)

Right of Professional Conscience

Engineers may use ethical professional judgment. Example: reject unsafe designs.

Right of Conscientious Refusal

Engineers may refuse unethical work. Examples: fake data, hide safety problems, misrepresent results.

Employer should reasonably accommodate such refusals.

Engineers and the Defense Industry

No universally correct answer — should I make weapons?

- Working in defense: ethically acceptable.
- Not working in defense: also ethically acceptable.

Decision should match personal values. Questions to ask: Is it consistent with my values? Is it defensive? Am I comfortable with the consequences?

Whistle-Blowing ★★

Reporting illegal or unethical activities to management or outside authorities. Purpose: protect public welfare.

By Destination

- **Internal:** Inside the organization.
- **External:** Media, government, law enforcement.

By Disclosure

- **Anonymous:** Identity hidden.
- **Acknowledged:** Identity known — stronger legally, greater personal risk.

Four Conditions for Whistle-Blowing ★★★★★

★EXAM LINE Remember the order: Need → Proximity → Capability → Last Resort

- **Need:** Serious preventable harm exists.
- **Proximity:** Firsthand knowledge — not rumors.
- **Capability:** Reasonable chance of success.
- **Last Resort:** Internal methods already failed.

Permission vs. Obligation ★★★Very important exam point

Condition	Result
All four conditions satisfied	Permission — engineers may whistle-blow.
All four conditions satisfied + imminent serious harm	Obligation — engineer must whistle-blow.

Motives

Acceptable motives: protect public safety, stop ongoing harm, follow engineering ethics.

Unacceptable motives: revenge, personal grudge, money, fame.

★**EXAM LINE** *Always examine your motives before whistle-blowing.*

Preventing Whistle-Blowing

Companies should:

- Build ethical culture
- Encourage communication
- Protect employees from retaliation
- Admit mistakes

★**EXAM LINE** *Don't punish whistle-blowers.*

Goodrich A7-D Brake Case

- Unsafe brake design discovered.
- Management ignored engineer.
- Fake test report created.
- FBI investigated.
- Senate hearing followed.
- Brake rejected.

LESSON *Never falsify engineering data. Public safety > Company interests.*

Hartford Civic Center Roof Collapse

- Ignored warning signs
- Poor design
- Missing structural support
- Excessive trust in computer analysis

LESSON *Never ignore real-world evidence just because computer models say everything is fine.*

Comparison Table — Lecture 5

Topic	Key Idea
Confidentiality	Keep company secrets, even after leaving
Conflict of Interest	Personal interests affect judgment
Competitive Bidding	Compete honestly
Professional Conscience	Right to make ethical decisions
Conscientious Refusal	Right to refuse unethical work
Whistle-Blowing	Report wrongdoing to protect the public
Permission	May whistle-blow
Obligation	Must whistle-blow if serious imminent harm exists

Cases You Should Remember — Lecture 5

Case	Main Lesson
BART	Protect public safety even if employer disagrees
Goodrich Brake	Never falsify engineering data
Hartford Roof	Listen to real-world warning signs

Definitions — Lecture 5

Confidentiality: *Duty to keep employer/client information secret.*

Conflict of Interest: *Personal interests interfere (or appear to interfere) with professional judgment.*

Whistle-Blowing: *Reporting illegal or unethical actions to someone who can stop them.*

Professional Conscience: *Right to use ethical professional judgment.*

Conscientious Refusal: *Right to refuse unethical work.*

Lecture 6 — Risk and Safety

1. Definitions

Risk: *The possibility of harm or loss.*

Safety: *Freedom from damage, injury, or risk.*

★**EXAM LINE** *Nothing is 100% risk-free. Engineers should make designs as safe as reasonably possible.*

2. Main Idea of the Lecture

Safety is an engineering responsibility, not just a user responsibility. Good designs prevent small human mistakes from becoming disasters.

3. Risk Depends On...

People judge risk differently based on:

Lower Risk	Higher Risk
Voluntary	Involuntary
Temporary harm	Permanent harm
Low probability	High probability
Reversible	Irreversible
High exposure needed	Low exposure enough
Delayed harm	Immediate harm

These factors affect how dangerous people perceive a risk to be.

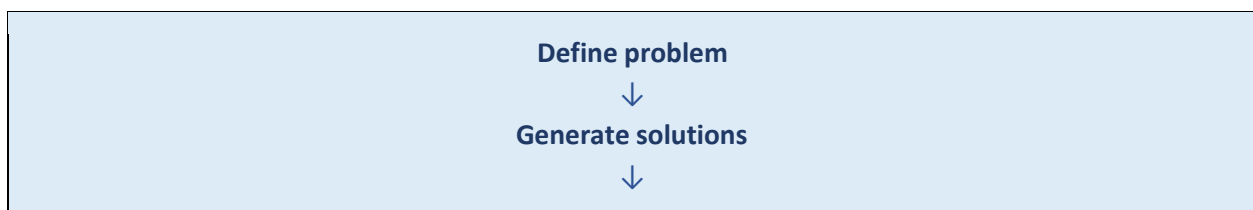
4. Four Criteria for Safe Design ★★★★★Write these in order

24. Legal compliance — meets laws and regulations.
25. Accepted engineering practice — follows professional engineering standards.
26. Safer alternatives explored — compare different designs before choosing one.
27. Foreseeable misuse addressed — design should remain safe even when users make predictable mistakes.

★**EXAM LINE** *A warning label is not a substitute for safer design.*

5. Safety in the Design Process

Safety should be considered during every stage.





KEY RULE *If safety conflicts with lower cost, safety should receive greater priority.*

6. Risk-Benefit Analysis

Compares benefits vs. risks to determine whether a project is worthwhile.

LIMITATION *Human life cannot be valued perfectly in money. Risk estimates are often uncertain.*

7. Environmental Racism

Definition: hazardous facilities (landfills, chemical plants, waste incinerators) are disproportionately located near poorer or minority communities.

Cause: cheap land, less political resistance.

★EXAM LINE *Benefits go to one group while risks fall on another.*

8. Three Types of Accidents ★★★★★

Procedural Accident

Cause: Human error, ignoring procedures.

Examples: missed inspection, wrong checklist.

Prevention: Training, supervision, better procedures.

Engineered Accident

Cause: Design flaw, material failure, poor testing.

Prevention: Better design, better testing, redesign.

Systemic Accident

Cause: Many small failures combine.

Characteristics: most difficult to detect; no single person causes it.

Prevention: Improve the entire system, not one part.

★EXAM LINE *Systemic accidents are the hardest to prevent.*

9. Engineering Duties

An engineer should:

- Follow engineering standards.
- Consider foreseeable misuse.

- Check all calculations before approval.
- Never sign work that has not been verified.
- Design for real-world use.
- Reduce risk whenever possible.

10. Case Studies — Lecture 6

ValuJet Flight 592 — Systemic

Cause: incorrect cargo handling, missing safety caps, poor packing, poor loading, design weaknesses.

LESSON *Many small mistakes can combine into one disaster.*

Hurricane Katrina — Systemic + Engineered

LESSON *Design critical structures for realistic hazards, not just historical ones.*

Firestone Tires / Ford Explorer — Engineered + Systemic

LESSON *Test the complete system, not just individual parts.*

Hyatt Regency Walkway Collapse — Engineered + Procedural

LESSON *Every design change must be fully recalculated before approval.*

Ford Crown Victoria — Engineered + Procedural

LESSON *Legal compliance alone is not enough if actual operating conditions are more dangerous.*

Teton Dam — Engineered + Procedural

LESSON *Never rely on one safety barrier.*

DC-10 Aircraft — Engineered + Procedural + Systemic

LESSON *Use fail-safe designs and thoroughly verify inspections.*

11. Patterns Across Cases ★

Case	Main Lesson
ValuJet	Small failures combine into disaster
Katrina	Design for realistic hazards
Firestone	Test systems, not only parts
Hyatt	Recheck every design change
Crown Victoria	Legal compliance isn't enough
Teton Dam	Multiple safety barriers are needed
DC-10	Fail-safe design and proper inspection

Memorize These One-Liners — Lecture 6

- Nothing is 100% risk-free.
- Safety is a design responsibility.

- Warning labels cannot replace safe design.
- Legal compliance is only the minimum requirement.
- Systemic accidents result from many small failures.
- Good engineering anticipates misuse.
- Design for real-world conditions, not ideal conditions.
- Test the entire system, not just individual components.

Lecture 7 — Environmental, Computer & Research Ethics

1. Therac-25 Case ★★★★★

What happened?

- Radiation therapy machine.
- Software bug caused massive radiation overdose.
- Operator saw “Malfunction 54”, but the manual didn't explain it.
- Machine falsely reported underdose.
- Operator resumed treatment.
- Patient had already received a deadly overdose.

Why did it happen?

- Software bug
- No hardware safety interlock
- Poor testing
- Poor documentation
- Confusing error messages
- Operator couldn't understand the error

ETHICAL LESSONS *Software errors can cause physical harm. Safety should never rely only on software. Systems should anticipate human mistakes. Error messages must be understandable. Thorough testing is essential.*

2. Three Areas of Engineering Ethics ★★

Area	Main Question
Environmental Ethics	How do engineers protect people and nature?
Computer Ethics	How should computers be used ethically?
Ethics in Research	How should research be conducted honestly?

3. Sustainable Design ★★★★★

Definition: designing products that protect both the environment and future generations.

Goals:

- Reduce pollution
- Consider manufacturing
- Consider disposal
- Maintain environmental quality
- Improve quality of life

4. Moral Standing ★★

Anthropocentric View: Nature matters because humans benefit. Example: protect forests because humans need them.

Broader View: Nature has value by itself. Animals, plants, and ecosystems deserve protection.

5. Two Environmental Approaches ★★★★★

Cost-Oblivious	Cost-Benefit
Environment first	Balance environment and cost
Rights & Duty Ethics	Utilitarianism
Difficult because “clean enough” is unclear	Difficult because life and nature cannot be priced easily

JUSTICE PROBLEM *Always ask: who gets the benefits? Who suffers the pollution?*

6. Albuquerque–Isleta Pueblo Case ★★

- Albuquerque followed EPA law.
- Isleta Pueblo wanted cleaner water for religious use.
- EPA supported Isleta.

LESSON *Legal compliance is not always ethically sufficient.*

7. Environmental Responsibility ★★★★★

Engineers should:

- Follow laws
- Protect public safety
- Protect environment
- Report harmful environmental effects

Also — know your limits. Ask experts when necessary.

8. Computer Ethics ★★★★★

Category 1 — Computer used to commit unethical acts

- Hacking
- Theft
- Privacy invasion
- Copyright violation

Category 2 — Computer used improperly as an engineering tool

- Wrong software
- Wrong simulation
- Unsafe embedded software

★EXAM LINE *The engineer remains responsible.*

9. Privacy, Hacking, Copyright ★★

Privacy: *People control access to their personal information.*

Hacking: *Unauthorized access is unethical, even without bad intentions.*

Copyright: *Easy copying does not remove ownership.*

10. Computers as Design Tools ★★

CAD, CAM, circuit analysis, Finite Element Analysis, structural analysis.

Benefits: Faster, more accurate.

Risks: Software bugs, wrong software, false confidence.

Engineers should know software limits, update software, and verify results manually.

KEY RULE *Software never replaces engineering judgment.*

11. Computers Inside Engineering Systems ★★

Examples: cars, aircraft, medical devices.

Problems: software failure, poor warning, human can't intervene.

Engineers should: test software, allow human intervention, use hardware redundancy.

12. Autonomous Computers ★★

Definition: computer makes decisions without humans.

Examples: manufacturing, stock trading, military weapons.

Problems: small errors can become disasters.

KEY RULE *Serious systems require meaningful human control.*

13. Computer Codes of Ethics ★★

Purpose: guide ethical decisions, encourage proper computer use.

EXAM LINE *Codes help, but cannot replace human judgment.*

14. Therac-25 Engineering Failures — Recap ★★

Remember these four points:

28. Software bug
29. Missing hardware interlock
30. Poor communication
31. Poor testing and documentation

15. Assigning Responsibility ★★

Engineers should design systems that:

- Expect user mistakes
- Prevent dangerous actions
- Give understandable warnings

Before returning a machine to service, engineers need: bug fixed, extensive testing, independent verification, better documentation, operator training.

16. Avanti vs. Cadence ★★

Employees may carry: Skills, experience, professional judgment.

Employees may NOT carry: Source code, trade secrets, confidential information.

17. Ethics in Research ★★★★★

During research: stay objective, change hypothesis if evidence changes, avoid bias.

When publishing: report honestly, don't exaggerate, give proper credit, correct mistakes.

KEY RULE *Honesty is the most important virtue in research.*

18. Pathological Science ★★★

Meaning: researchers unintentionally fool themselves.

Signs: weak evidence, extraordinary claims, ignores criticism, results cannot be confirmed.

Solution: independent verification, strong controls, repeat experiments.

19. N-Ray Case ★★★

LESSON *Expectation can influence observation.*

Scientists should repeat experiments, accept criticism, and verify objectively.

20. Cold Fusion Case ★★★

Problems: results couldn't be reproduced, possible contamination, weak experimental controls.

★EXAM LINE *Investigate evidence against your theory as carefully as evidence supporting it.*

21. Ghostwriting ★★★

Wrong because: the real writer gets no credit, the fake author gets credit, readers can't detect conflicts of interest.

★EXAM LINE *Everyone who contributed significantly should receive proper credit.*

22. Ten One-Line Answers — Lecture 7

- **Sustainable Design:** Designing products that protect the environment throughout their life cycle.
- **Anthropocentric View:** Nature is valuable because it benefits humans.
- **Moral Standing:** Who deserves ethical consideration.
- **Hardware Redundancy:** Backup hardware that keeps a system safe if software fails.
- **Autonomous Computer:** Computer that makes decisions without human intervention.
- **Privacy:** Control over personal information.
- **Hacking:** Unauthorized access to a computer system.
- **Pathological Science:** Researchers unintentionally deceive themselves.
- **Ghostwriting:** Someone writes research but another person receives authorship.
- **Therac-25 Lesson:** Never rely only on software for safety-critical systems.

Appendix — NSPE Code of Ethics

Purpose

The NSPE Code of Ethics provides ethical guidelines for engineers. It emphasizes:

- Protecting public safety.
- Maintaining honesty and integrity.
- Working within competence.
- Serving clients ethically.
- Protecting the environment.
- Respecting fellow engineers.

Golden Rule ★★★★★Most important

KEY RULE *Engineers shall hold paramount the safety, health, and welfare of the public.*

Priority order:

32. Public
33. Client / Employer
34. Personal interest

★EXAM LINE *If these conflict, public safety always wins.*

Six Fundamental Canons

Canon	Meaning
1	Protect public safety, health, and welfare.
2	Work only within your competence.
3	Make objective and truthful statements.
4	Be a faithful agent or trustee for clients/employers.
5	Avoid deceptive acts.
6	Behave honorably, responsibly, ethically, and lawfully.

Rules of Practice

Public Safety

- Report dangerous situations.
- Approve only standard-compliant designs.
- Report ethical violations.
- Don't help illegal engineering practice.
- Don't associate with fraudulent organizations.

Competence

Engineers should:

- Accept work only if qualified.

- Sign only work they understand.
- Coordinate large projects only if each technical section is signed by the qualified engineer.

Public Statements

Must be truthful, objective, fact-based, complete. If paid to express an opinion, disclose that relationship.

Employer & Client

Engineers must:

- Avoid conflicts of interest.
- Keep confidential information secret.
- Not accept multiple payments without everyone's consent.
- Not accept gifts or bribes.
- Government engineers must avoid decisions involving their own organizations.

Avoid Deception

Never:

- Fake qualifications.
- Exaggerate experience.
- Offer or accept bribes.
- Misrepresent achievements.
- Mislead clients or the public.

Professional Obligations

Honesty

- Admit mistakes.
- Don't change facts.
- Warn clients if projects are likely to fail.
- Don't misuse outside employment.

Public Interest

Engineers should:

- Help society.
- Guide students.
- Promote engineering.
- Follow sustainable development.
- Withdraw from unethical projects if necessary.

Confidentiality

Never disclose business information, technical processes, or client secrets — unless legally required or permitted.

Conflicts of Interest

Don't let money, gifts, commissions, or supplier benefits affect engineering decisions.

Respect Other Engineers

Don't spread false criticism, steal credit, or damage reputations. Report genuine unethical conduct properly.

Intellectual Property

Give proper credit. Respect patents, copyrights, designs, and ownership agreements. Employer-created engineering records usually belong to the employer.

Professional Development

Continue learning through courses, seminars, technical literature, and professional practice.

Sustainable Development

Sustainable Development: *Meeting today's needs without harming the ability of future generations to meet theirs.*

Engineers are encouraged to: conserve natural resources, reduce pollution, protect the environment.

Other Organizations — Know the Differences

Organization	Main Focus
American Institute of Chemical Engineers	Human welfare, no harassment, protect environment
American Society of Civil Engineers	Sustainable development
American Society of Mechanical Engineers	Human welfare
Association for Computing Machinery	Public interest first, privacy, software quality
Institute of Electrical and Electronics Engineers	Public safety, disclose risks, fairness
Institute of Industrial Engineers	Integrity and human welfare

Frequently Tested Definitions

Faithful Agent (Trustee): *Acts honestly and loyally for the employer/client while placing public safety above everything else.*

Conflict of Interest: *A personal or financial interest that may affect professional judgment. Example: accepting gifts from a supplier whose product you're evaluating.*

Competence: *Only performing work you are qualified to do through education or experience.*

Deceptive Act: *Any dishonest action, such as lying, hiding facts, fake qualifications, bribery, or misleading advertisements.*

Confidential Information: *Information that belongs to a client or employer and cannot be shared without permission.*

Sustainable Development: *Development that satisfies present needs without compromising future generations.*

— End of Cheat Sheet —